Polarized by Design

1 Model Description

1.1 Opinions

Let the social network be an undirected graph $G = (V, E)$ with $|V| = N$ nodes. At each discrete time $t \in \mathbb{N}_0$, every node $i \in V$ holds one of two opinions,

$$x_i(t) \in \{F, D\}, \tag{1}$$

where F (Fake-believer) and D (Debunker) denote the two competing stances. All nodes update their opinions synchronously at every time step.

1.2 Stubbornness as a node property

Each node i is endowed with a *stubbornness* parameter

$$S_i \in [1,\, 100],$$

which quantifies its resistance to external influence. A higher value means a stronger adherence to the current opinion; $S_i = 100$ corresponds to practical immovability, while $S_i = 1$ represents minimal resistance.

1.3 Single-neighbor conversion probability

Consider a pair of adjacent nodes $(i, j) \in E$ with $x_i(t) \neq x_j(t)$. A persuasion attempt from j toward i succeeds if

$$S_i\, r < w_{ij}(t), \tag{2}$$

where $r \sim \mathcal{U}(0, 1)$ is a random number drawn afresh for each attempt, and $w_{ij}(t) \in [0, 1]$ is the current weight (trust) of the edge ij. Condition (2) may be interpreted as follows: the friendship weight w_{ij} must overcome the node's stubbornness S_i, up to a random fluctuation, for the opinion change to occur.

Because r is uniformly distributed on $(0, 1)$,

$$\Pr(S_i r < w_{ij}) \;=\; \Pr\Big(r < \frac{w_{ij}}{S_i}\Big) \;=\; \frac{w_{ij}}{S_i},$$

so that the per-contact conversion probability is simply $p_{ij}(t) = \dfrac{w_{ij}(t)}{S_i}$.

1.4 Combined effect of multiple disagreeing neighbours

Node i may have several neighbours who hold the opposite opinion. We assume that those neighbours act independently, and that a single successful persuasion attempt suffices to flip i's opinion. Thus i remains in its current state only if *every* disagreeing neighbour fails. The probability that neighbour j fails is $1 - \frac{w_{ij}}{S_i}$; hence the probability that all disagreeing neighbours fail simultaneously is

$$\prod_{\substack{j \in \mathcal{N}(i) \\ x_j(t) \neq x_i(t)}} \Big(1 - \frac{w_{ij}(t)}{S_i}\Big),$$

where $\mathcal{N}(i)$ denotes the set of neighbours of i. Consequently, the probability that *at least one* neighbour succeeds, i.e. that node i flips its opinion, is

$$\boxed{p_{\text{flip}}(i;t) = 1 - \prod_{\substack{j \in \mathcal{N}(i) \\ x_j(t) \neq x_i(t)}} \Big(1 - \frac{w_{ij}(t)}{S_i}\Big)}. \tag{3}$$

Equation (3) compactly captures the aggregate influence of the entire neighbourhood.

1.5 The state update rule

We do **not** simulate each neighbour's attempt separately. The product formula compresses the entire neighbourhood into a single conversion probability. For each node i at every time step:

1. **Compute** the combined conversion probability $p_{\text{flip}}(i)$ using the product formula above.
2. **Draw a single uniform random number** $r \sim \mathcal{U}(0,1)$.
3. **If** $r < p_{\text{flip}}(i)$, the node switches to the opposite opinion; **otherwise** – including the case $r = p_{\text{flip}}(i)$ – the node stays as it was.

For a node that is currently a Fake-believer, the rule written in equation form is

$$\boxed{x_i(t+1) = \begin{cases} D, & \text{if } r < 1 - \displaystyle\prod_{\substack{j \in \mathcal{N}(i) \\ x_j(t) = D}} \Big(1 - \frac{w_{ij}(t)}{S_i}\Big), \\ F, & \text{otherwise} \end{cases} \qquad r \sim \mathcal{U}(0,1).}. \tag{4}$$

The symmetric rule applies to a node that is currently a Debunker. Because there are only two states, a node can change at most once per step – no extra dice rolls are needed.

1.6 Adaptive weights: Hebbian update

After all opinions have been updated, edge weights are adjusted according to a constant-step Hebbian rule:

$$w_{ij}(t+1) = \Pi_{[0,1]}\Big(w_{ij}(t) + \Delta_{ij}(t)\Big), \tag{5}$$

where the weight change is determined by the (dis)agreement of the two endpoints,

$$\Delta_{ij}(t) = \begin{cases} +a, & \text{if } x_i(t) = x_j(t), \\ -a, & \text{if } x_i(t) \neq x_j(t), \end{cases}$$

and $\Pi_{[0,1]}$ denotes the projection onto the unit interval,

$$\Pi_{[0,1]}(x) = \min\big(1,\, \max(0, x)\big) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

The parameter $a > 0$ is the *step size*. Agreement reinforces a tie, whereas disagreement erodes it.

The interplay between persuasion and weight decay creates a dynamical race: if persuasion occurs quickly, a cross-cutting edge may turn into an agreeing edge before it is weakened too much; if persuasion is too slow, the weight eventually drops to zero, permanently insulating the two individuals from each other.

1.7 Initial conditions

At $t = 0$, the system is initialised as follows:

1. **Network:** Generate a synthetic random graph (Erdős–Rényi, Watts–Strogatz, or Barabási–Albert) with N nodes and a specified average degree $\langle k \rangle$.
2. **Opinions:** Assign each node F or D independently with equal probability $\frac{1}{2}$, so that the initial opinion distribution is maximally mixed.
3. **Weights:** Set $w_{ij}(0) = w_0$ for every existing edge $(i, j) \in E$, where $w_0 \in [0, 1]$ is a constant (e.g. $w_0 = 0.5$).
4. **Stubbornness:** Assign a stubbornness value $S_i \in [1, 100]$ to each node, either a global constant S or drawn from a prescribed distribution. In our initial model, we keep this constant, representing teh average stubbornness of the population.

The dynamics (3)–(5) are then iterated for T time steps, and the evolution of the system is monitored.

1.8 Polarization coefficient

To quantify the degree of polarization, we introduce a measure based on the weights of edges that connect opposite opinions. Let the set of *cross-opinion* edges at time t be

$$E_\times(t) := \{(i, j) \in E \mid x_i(t) \neq x_j(t)\}.$$

The average weight on those edges is

$$\bar{w}_\times(t) := \begin{cases} \dfrac{1}{|E_\times(t)|} \displaystyle\sum_{(i,j) \in E_\times(t)} w_{ij}(t), & \text{if } |E_\times(t)| > 0, \\ 0, & \text{if } |E_\times(t)| = 0. \end{cases} \tag{6}$$

Since every $w_{ij} \in [0, 1]$, we have $0 \leq \bar{w}_{\times}(t) \leq 1$. The *polarization coefficient* is then defined as

$$P(t) := 1 - \bar{w}_{\times}(t). \tag{7}$$

It follows that $0 \leq P(t) \leq 1$. Low values of $P(t)$ indicate that cross-opinion ties are still substantial (low polarization), whereas values close to 1 signify that those ties have essentially vanished (high polarization). The extremal case $P(t) = 1$ occurs precisely when no cross-opinion edges exist, i.e., when the network has fragmented into mutually unreachable opinion clusters.

1.9 Symbol table

Symbol	Type	Meaning
$x_i(t)$	variable	opinion of node i at time t: F or D
$w_{ij}(t)$	variable	weight of edge ij at time t ($0 \leq w_{ij} \leq 1$)
S_i	parameter	stubbornness of node i ($1 \leq S_i \leq 100$)
a	parameter	weight step size after agreement or disagreement
w_0	parameter	initial edge weight
$P(t)$	observable	polarization coefficient (0 = unpolarized, 1 = fully fragmented)